NATIONAL ENVIRONMENT MANAGEMENT AUTHORITY (NEMA)

CALL FOR PROPOSALS FOR INNOVATIONS STUDENT SUPPORT GRANT 2026–2027

APPLICATION FORM

Eligibility: *The innovations student support grant is intended to support environmental innovations at piloting phase. Proposals focused only on idea generation, basic research, concept development or initial prototype development will* ***NOT*** *be considered.*

1. APPLICANT AND INSTITUTIONAL DETAILS

Full Name	WASSWA FRANCIS
National ID/Passport No.	CM00093106P55A
Name of University	BUSITEMA UNIVERSITY
Faculty/School/College	ENGINEERING AND TECHNOLOGY
Department	CHEMICAL AND PROCESS ENGINEERING
Programme	BACHELOR OF SCIENCE IN AGRO-PROCESS ENGINEERING
Student Registration No.	BU/UG/2024/2669
Email Address	franciswasswa329@gmail.com
Telephone Number	+256 758797524
Supervisor's Name	Mr. MASERUKA BENEDICTO
Supervisor's Contact	+256 770365458, mbendicto.eng@busitema.ac.ug

Supervisor's Name: Mr. MASERUKA BENDICTO

Supervisor's Contact: +256 785700880

2. PROPOSED ENVIRONMENTAL INNOVATION

2.1 Title of the proposed innovation:

AN AGRICULTURAL WASTE VALORIZATION AND VALUE-ADDITION TECHNOLOGY THAT CONVERTS LIGNOCELLULOSIC AGRICULTURAL RESIDUES INTO FERMENTABLE SUGARS USING AN IMMOBILIZED-ENZYME PACKED-BED REACTOR.

2.2 Environmental priority area (Indicate as per the call)

Primary area: A. Pollution and Waste Management. The innovation focuses on converting agricultural/agro-processing lignocellulosic residues into useful fermentable sugars, supporting waste-to-resource approaches, circular economy and resource efficiency.

3. RESEARCH ORIGIN OF THE INNOVATION

Briefly describe the research study from which the innovation emerged, including key findings (Max. 300 words)

The innovation emerged from the applicant's undergraduate work in agro-processing engineering and interest in developing practical ways of recovering value from agricultural residues. The proposed study brings together enzyme technology, reaction engineering and process modelling to address the underutilization of lignocellulosic biomass.

Previous research has shown that cellulolytic enzymes can hydrolyse cellulose and that immobilizing enzymes can improve operational stability and make continuous processing in packed-bed reactors possible. However, practical performance is affected by enzyme loading, temperature, pH, substrate concentration, residence time, mass transfer and enzyme deactivation. These findings provide the scientific basis for developing and testing a continuous packed-bed system.

The proposed innovation will adapt these established concepts to a selected locally available lignocellulosic residue. The work will determine suitable pretreatment and operating conditions, quantify sugar release, develop a kinetic model, and use the experimental results to assess reactor performance, and scale-up requirements. The expected outcome is a practical pilot-ready system that demonstrates how an agricultural residue can be changed from a low-value waste stream into a useful carbohydrate resource.

4. ENVIRONMENTAL MANAGEMENT CHALLENGE

Describe the specific environmental management challenge the innovation seeks to address. (Max. 200 words)

Agricultural and agro-processing activities generate large quantities of lignocellulosic residues containing cellulose and hemicellulose that are often underutilized. When these residues are poorly managed or disposed of without value recovery, they occupy space and can contribute to local environmental burdens.

At the same time, the carbohydrate fraction in these residues represents a potentially useful raw material for producing fermentable sugars and other bio-based products. A major challenge is therefore not simply the presence of the waste, but the lack of practical, efficient and potentially scalable technologies for converting it into a useful resource.

This innovation addresses that gap by developing a continuous packed-bed enzymatic hydrolysis system. The pilot will demonstrate whether selected lignocellulosic waste can be efficiently converted into fermentable sugars while monitoring material use, energy requirements, conversion, productivity and operating cost. The approach supports waste minimization, resource recovery and circular bioeconomy principles.

5. DESCRIPTION OF THE INNOVATION

Describe the novelty of the innovation. (Max. 200 words)

The novelty of the innovation lies in integrating immobilized-enzyme packed-bed hydrolysis with experimental optimization, kinetic modelling and preliminary scale-up and economic assessment in one practical system. Rather than treating lignocellulosic waste only as a disposal problem, the reactor is designed to recover its carbohydrate fraction as fermentable sugars.

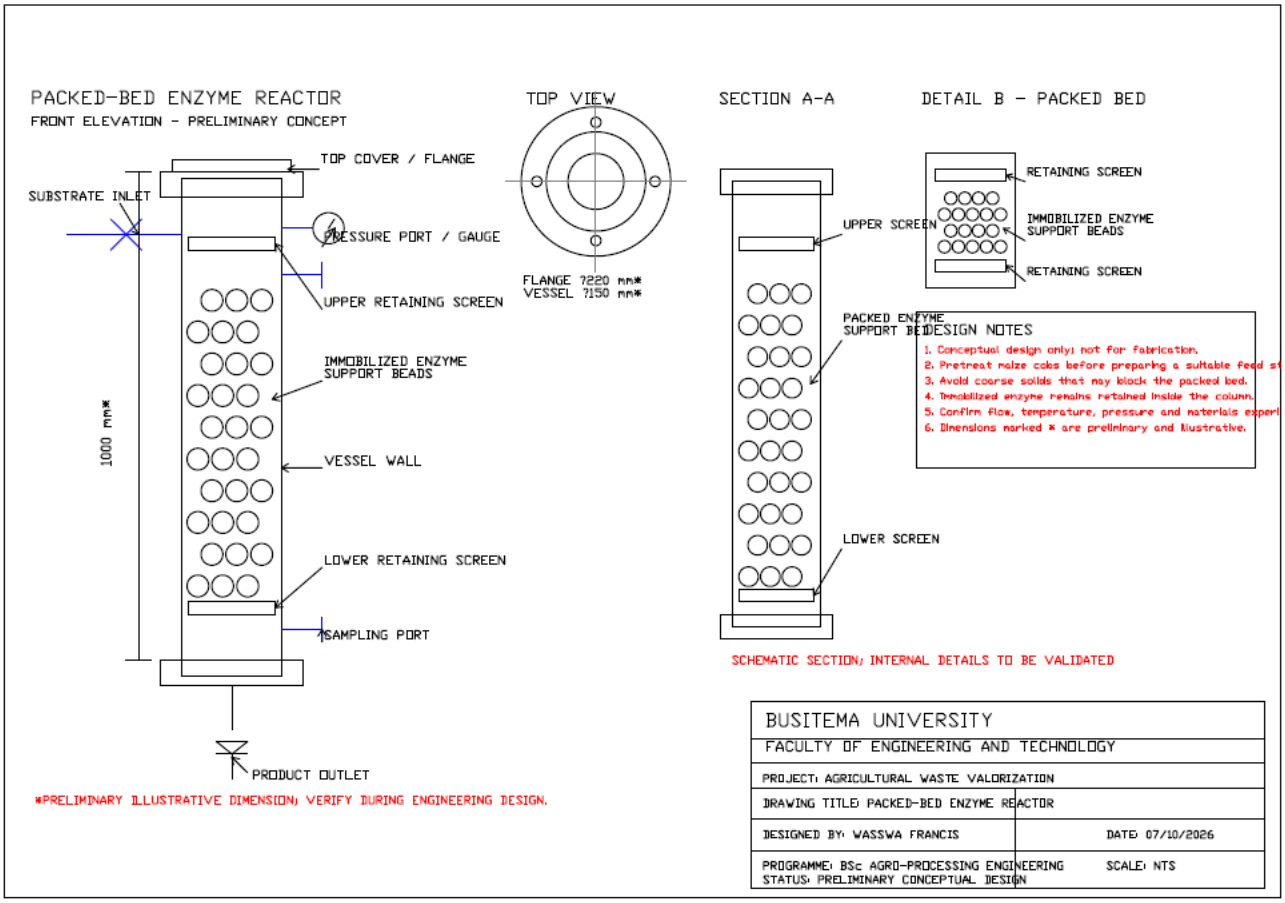
The packed-bed configuration is intended to allow continuous operation and repeated use of the immobilized enzyme, reducing the need to add fresh enzyme for every batch. The study will also use experimental data to develop a kinetic model that can predict reactor behaviour under different flow rates and operating conditions.

The innovation will be tested as a physical pilot system rather than remaining at the idea or simulation stage. Its performance will be assessed using measurable indicators such as sugar yield, conversion, productivity, enzyme stability, residence time and resource consumption. The resulting data will support decisions about replication and scale-up.

6. EVIDENCE OF COMPLETED IDEATION AND PROTOTYPING

Provide clear evidence that the innovation has completed ideation and prototyping, such as a prototype, proof of concept, laboratory validation, demonstration results, technical drawings, software/platform, photographs or other relevant evidence. (Max. 200 words)

The idea emerged from the need to transform underutilized agricultural residues into valuable industrial raw materials. In Uganda, maize cobs offer a renewable feedstock for producing fermentable sugars, although lignocellulose's complex structure limits efficient conversion. The proposed innovation involves developing a laboratory-scale packed-bed reactor containing immobilized enzymes to hydrolyse pretreated maize cobs into soluble sugars for fermentation and other biotechnological applications. The concept integrates biomass pretreatment, enzyme immobilization, reactor design, process optimization, and kinetic modelling. The next phase will involve laboratory testing, prototype fabrication, and evaluation of sugar yield, enzyme reusability, flow behaviour, and operational feasibility.



The attached illustration presents the preliminary conceptual design of the proposed immobilized-enzyme packed-bed reactor. The design is informed by published research on cellulase immobilization and established biomass characterization methods (Xu et al., 2023; Sluiter et al., 2012). The proposed system is intended to investigate the conversion of pretreated maize-cob

biomass into fermentable sugars. Further engineering design, experimental testing and performance validation will be required to establish its operating conditions and feasibility. The illustration represents a conceptual design rather than a fabricated or experimentally validated prototype.

7. PILOT OBJECTIVE, SITE AND INTENDED USERS/BENEFICIARIES

Clearly illustrate the pilot objectives, where the pilot will take place, and the intended users/beneficiaries (Max. 250 words)

The proposed six-month pilot will demonstrate continuous enzymatic hydrolysis of a locally available lignocellulosic agricultural or agro-processing residue using a packed-bed reactor containing immobilized hydrolytic enzymes. The objectives are to establish stable reactor operation, determine suitable operating conditions, measure cellulose conversion and fermentable-sugar yield, evaluate enzyme stability and reactor productivity, and generate data for kinetic modelling, scale-up, and techno-economic assessment.

The pilot will be conducted at Busitema University using available laboratory facilities, with additional analytical or fabrication support where necessary. The primary beneficiaries will be students, researchers, and academic staff in agro-processing, chemical and process engineering, biotechnology, and environmental management.

The project may also benefit agro-processors and research institutions interested in converting agricultural residues into valuable products, including fermentable sugars for biofuel production. This initiative will promote agricultural waste utilization and sustainable resource management.

To ensure feasibility within six months, the pilot will focus on one selected feedstock, enabling effective process demonstration, reliable performance evaluation, and generation of data to support future research and industrial applications.

8. EXPECTED RESULTS AND INDICATORS

No.	Expected result	Indicator	Target	Means of verification
1	Working pilot reactor demonstrated	Continuous operation achieved without major leakage/blockage	At least 3 successful continuous runs	Run sheets, photographs, commissioning records
2	Lignocellulosic waste converted	Biomass mass balance and conversion	Baseline established and measurable conversion achieved	Mass-balance records and laboratory results
3	Fermentable sugars produced	Sugar concentration/yield	Quantified sugar yield under optimized conditions	Laboratory analysis and calibration records

4	Operating conditions optimized	Temperature, pH, enzyme loading and residence time	An experimentally supported operating window identified	Experimental dataset and statistical analysis
5	Kinetic model developed and validated	Model fit/error against experimental data	Model predictions agree acceptably with validation data	MATLAB files, plots and validation results

9. ADOPTION, REPLICATION AND SCALE-UP

Demonstrate how the innovation could be adopted, replicated or scaled if the pilot is successful. Identify potential adopters/partners and conditions required for scale-up. (Max. 200 words)

If the pilot demonstrates reliable conversion and acceptable resource and economic performance, the reactor can be replicated using locally available materials and scaled by increasing reactor volume, bed capacity and flow throughput while maintaining appropriate residence time and operating conditions.

Potential adopters or partners include agro-processing companies generating maize, wheat, sugarcane or other lignocellulosic residues; universities and research laboratories; biotechnology and bio-based product enterprises; and organisations working on circular economy and waste-to-resource technologies.

Scale-up will require a consistent feedstock supply, reliable pretreatment, stable enzyme immobilization, suitable pumps and process controls, analytical capacity, and evidence that the cost of enzyme use and other inputs is acceptable. The kinetic model and pilot data will provide the engineering basis for sizing larger systems. Partnerships with agro-processors can also provide waste feedstock and a pathway for technology demonstration and eventual adoption.

10. WORK PLAN.

No.	Key pilot activity	Expected output	Duration
1	Finalize pilot design, select feedstock and prepare protocols	Final reactor design, feedstock and test protocols	Month 1
2	Fabricate/assemble and commission pilot reactor; prepare enzyme and substrate	Assembled and commissioned pilot system	Month 1–2
3	Conduct baseline and operating-condition experiments	Experimental performance dataset	Month 3
4	Optimize hydrolysis conditions and evaluate enzyme/reactor stability	Optimized operating window and stability results	Month 4

5	Develop and validate kinetic model; assess scale-up parameters	Validated model and preliminary scale-up calculations	Month 5
6	Demonstration and reporting	Pilot demonstration, final analysis and dissemination report	Month 6

List key activities for implementing the innovation. The proposed pilot should be completed within 6 months.

11. BUDGET

Budget should focus on piloting, field testing, validation, demonstration and documentation. (*Maximum support amount is Uganda shillings 7,000,000*)

Proposed pilot budget (maximum support: UGX 7,000,000):

No.	Budget item/activity	Unit	Quantity	Unit cost (UGX)	Total cost (UGX)
1	Pilot reactor fabrication materials, fittings and assembly	Lot	1	1,200,000	1,200,000
2	Pump, flow-control components, tubing and connectors for pilot operation	Lot	1	850,000	850,000
3	Enzymes, immobilization materials and laboratory consumables	Lot	1	1,000,000	1,000,000
4	Feedstock preparation and pretreatment chemicals/consumables	Lot	1	500,000	500,000
5	Laboratory analyses and sample testing	Lot	1	1,000,000	1,000,000
6	Pilot testing, field/sample collection and local transport	Lot	1	450,000	450,000
	Data analysis, modelling, validation and documentation	Lot	1	350,000	350,000
	Pilot demonstration, dissemination and reporting	Lot	1	400,000	400,000
	Contingency for essential pilot consumables and fittings	Lot	1	500,000	500,000

12. REQUIRED SUPPORTING DOCUMENTS

Attach the following documents in a single PDF file.

i) Curriculum Vitae of the Innovator (Max. 2 pages)
ii) Signed Academic Testimonial/ provisional results (as proof of student status)
iii) Letter of recommendation from the Head of Department addressed to ED, NEMA.
iv) Letter of Expression of Interest addressed to ED, NEMA.
v) Copy of National ID/Passport
vi) Declaration form (Appendix 1)

CURRICULUM VITAE

WASSWA FRANCIS

BSc Agro-Processing Engineering Student

Busitema University, Faculty of Engineering and technology | Uganda

REG.NO: BU/UG/2024/2669

Telephone: +256 758 797524 | Email: franciswasswa329@gmail.com

PROFESSIONAL PROFILE

Undergraduate Agro-Processing Engineering student at Busitema University with industrial exposure to wheat milling, production operations and quality control. Interested in sustainable processing, agricultural waste valorization and value-addition technologies. Developing a proposal-stage innovation to convert maize cobs into fermentable sugars using an immobilized-enzyme packed-bed reactor.

EDUCATION AND ACADEMIC QUALIFICATIONS

INSTITUTION/SCHOOL	QUALIFICATION	YEAR
Busitema University, Faculty of Engineering and Technology	Bachelor of science in Agro-Processing engineering	Expected graduation: 2028 Current results; CGPA: 4.05
Westville high school-Fort portal	Uganda Advanced Certificate of Education (UACE)	2023
St. John the Baptsit-Ndibatta	Uganda Certificate of Education (UCE	2021
Trinity parents school-Ssumbwe	Primary Leaving Examination (PLE)	2016

INNOVATION / RESEARCH PROJECT

An Agricultural Waste Valorization and Value-Addition Technology That Converts Lignocellulosic Agricultural Residues into Fermentable Sugars Using an Immobilized-Enzyme Packed-Bed Reactor

- Project stage: Concept and proposal development; maize cobs selected as the proposed feedstock.
- Proposes a process involving feedstock preparation, pretreatment, enzymatic hydrolysis in an immobilized-enzyme packed bed, and assessment of sugar production.
- Planned work includes process optimization, kinetic modelling, scale-up assessment and preliminary techno-economic analysis.
- The reactor illustration/design is conceptual; experimental performance and prototype validation are yet to be established.

INDUSTRIAL TRAINING

Master Grain Milling Company Ltd. — Production and Quality Control

Industrial trainee | 08th/06/2026-07th/08/2026.

- Gained practical exposure to wheat cleaning, conditioning/tempering and milling operations.
- Observed production monitoring and quality-control activities in an industrial grain-processing environment.

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- Developed familiarity with moisture monitoring, flour and by-product sampling, and process documentation.

AWARDS AND ACHIEVEMENTS

- Second Place : Food Science Quiz Competition: Participated as a member of the Busitema University team that secured second place in a Food Science Quiz competition held at Kyambogo University on 25Th/09/2026.

TECHNICAL AND PROFESSIONAL SKILLS

- AutoCAD: basic engineering drawing and conceptual design.
- Microsoft Excel: data recording, tabulation and basic analysis.
- Industrial exposure: grain-processing operations, process monitoring and quality-control procedures.

REFEREES

1. Mr. Sserumaga Paul.

Position: Head of Department, Chemical and process engineering.

Institution: Busitema University, Faculty of Engineering and Technology.

Telephone: +256 772 311861/+256 700377368

Email:sserumagap@gmail.com

2. Mr. Jared Misati

Position: Head of Department, Production and quality control

Institution: Master Grain Milling Company LTD-Njeru

Telephone: +256 768103574

Email:jaredmi sati1983@gmail.com

3. Mr. Luyombya Benjamin.

Position: School director.

Institution: Westville high school-Fort Portal.

Telephone: +256 702 673955

Email:

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BUSITEMA UNIVERSITY

Office of the Academic Registrar
Jinja - Malaba Road P.O.BOX 236, Tororo
Tel: 0454448808
Email: vc.adm@busitema.ac.ug, Website: https://busitema.ac.ug/

BUSITEMA UNIVERSITY
ACADEMIC REGISTRAR'S OFFICE
22 OCT 2026
FACULTY OF ENGINEERING
REGISTERED

TESTIMONIAL

Print Date: Thu, Oct 22, 2026 9:39 AM



FULL NAME: **WASSWA Francis**
STUDENT NO.: **2400402669**
REG. NO.: **BU/UG/2024/2669**
GENDER: **MALE**
DATE OF BIRTH: **01/05/2002**
FACULTY: **FACULTY OF ENGINEERING AND TECHNOLOGY**
PROGRAMME: **(APE) BACHELOR OF SCIENCE IN AGRO-PROCESSING ENGINEERING**

YEAR 1 - 2024/2025 - SEMESTER I

CODE	TITLE	MARK	CUs	GRADE	GD POINT	REMARK
APE1111	ENGINEERING MATHEMATICS	78	4	B+	4.5	NP
APE1112	COMPUTER APPLICATIONS	77	3	B+	4.5	NP
APE1113	ENGINEERING MECHANICS I	83	4	A	5.0	NP
APE1114	ENGINEERING DRAWING	68	4	C+	3.5	NP
APE1115	COMMUNICATION SKILLS	70	3	B	4.0	NP
APE1116	ENGINEERING FOR SUSTAINABLE DEVELOPMENT	68	3	C+	3.5	NP
APE1117	BIOCHEMISTRY	66	3	C+	3.5	NP
SEMESTER REMARK: NP				**GPA: 4.10**		**CGPA: 4.10**

YEAR 1 - 2024/2025 - SEMESTER II

CODE	TITLE	MARK	CUs	GRADE	GD POINT	REMARK
APE1211	ENGINEERING MATHEMATICS II	79	4	B+	4.5	NP
APE1212	MECHANICS OF MATERIALS I	73	4	B	4.0	NP
APE1213	ELECTRICAL TECHNOLOGY	80	3	A	5.0	NP
APE1214	ENGINEERING MECHANICS II	55	4	D+	2.5	NP
APE1215	INTRODUCTION TO FOOD MICROBIOLOGY	73	3	B	4.0	NP
APE1216	COMPUTER AIDED DESIGN (CAD)	88	3	A	5.0	NP
SEMESTER REMARK: NP				**GPA: 4.10**		**CGPA: 4.10**

YEAR 2 - 2025/2026 - SEMESTER I

CODE	TITLE	MARK	CUs	GRADE	GD POINT	REMARK
APE2111	ENGINEERING MATHEMATICS III	84	4	A	5.0	NP
APE2112	MATERIAL SCIENCE	61	4	C	3.0	NP
APE2113	COMPUTER PROGRAMMING	75	4	B+	4.5	NP

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YEAR 1 - 2024/2025 - SEMESTER I

CODE	TITLE	MARK	CUs	GRADE	GD POINT	REMARK
APE2114	ELECTRICAL DEVICES AND MACHINES	68	3	C+	3.5	NP
APE2115	REACTION KINETICS	74	3	B	4.0	NP
APE2116	EXTENSION FOR ENGINEERS	66	3	C+	3.5	NP
SEMESTER REMARK: NP				**GPA: 3.95**		**CGPA: 4.05**

Minimum Graduation Load(CUs): 179
Total Credit Units: 66

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Dean's Signature

NOTE: THE RESULTS ON THIS DOCUMENT ARE PROVISIONAL, AND MUST BEAR A STAMP FROM THE REGISTRAR. A FULL TRANSCRIPT IS ISSUED BY THE ACADEMIC REGISTRAR.
THE MEDIUM OF INSTRUCTION IS ENGLISH.

NORMAL PROGRESS: NP
RETAKE PAPERS: RT
CONCEDED PASS: CP
AUDITED COURSE: (AU)

This is a Testimonial as at Oct 22, 2026 and is Valid with a stamp and signature. Any alterations of this document makes it invalid

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BUSITEMA UNIVERSITY
Pursuing Excellence
Department of Chemical and Process Engineering

P.O. Box 236, Tororo, Uganda
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Fax: +256 - 45 4436517
Email: info@adm.busitema.ac.ug
www.busitema.ac.ug

October 21st, 2026

The Executive Director,
National Environment Management Authority,
Kampala, Uganda.

Dear Sir/ Madam,

Re: Recommendation of Mr. Wasswa Francis for your innovation challenge

This is to recommend Mr. Wasswa Francis for the NEMA Innovation Grant under Circular Bio-economy and waste management.

Francis is an enthusiastic and capable engineering student engaged on a BSc. Agro-Processing Engineering course. The course has exposed them to principles and practice of environmental Engineering.

During this call, Francis intends to Develop and Optimise a packed-bed enzymatic system for converting agricultural lignocellulosic waste into fermentable sugars for bio-fuel production. I know he is capable to pull off this project. I also pledge on behalf of the Department to give hive technical backstopping and disposal to research facilities including laboratories and the fabrication workshop.

Yours sincerely,

BUSITEMA UNIVERSITY
DEPARTMENT OF CHEMICAL AND PROCESS ENGINEERING
21 OCT 2026
FACULTY OF ENGINEERING AND TECHNOLOGY

SSERUMAGA Paul
HoD., CHEMICAL & PROCESS ENGINEERING DEPARTMENT.
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Cell: +256700-377368

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FACULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF CHEMICAL AND PROCESS ENGINEERING

LETTER OF EXPRESSION OF INTEREST

"Development and Optimization of a Packed-Bed Enzymatic System for Converting Agricultural Lignocellulose Waste into Fermentable Sugars"

by

WASSWA FRANCIS

BU/UG/2024/2669

Tel: +256 758797524 | Email: fwasswa242669@busitema.ac.ug

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Date: 08, OCT, 2026

The Executive Director

National Environment Management Authority (NEMA)

Kampala, Uganda

Dear Sir/Madam,

Re: NEMA Innovation Grant Application - Circular Bioeconomy & Waste Valorization.

The National Environment Management Authority (NEMA) has identified waste management, environmental protection, and sustainable resource utilization as critical priorities for Uganda development agenda. In this spirit, I am writing to express my strong interest in the NEMA Innovation Grant program and to introduce a transformative research initiative that directly aligns with these priorities.

I propose to develop and optimize a packed-bed enzymatic system for converting agricultural lignocellulose waste into fermentable sugars, underpinned by circular bioeconomy principles. This project addresses one of Uganda most pressing environmental challenges: the management and valorization of agricultural residues.

This project aligns academic research with environmental action. I am proposing to solve a critical challenge affecting Uganda air quality, climate, soil health, and economic productivity.

I respectfully request NEMA consideration and look forward to partnership in this transformative initiative.

Yours faithfully,

WASSWA FRANCIS

Third Year Undergraduate Student



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NIRA
National Identification & Registration Authority

FORM 2
Regulation 8 (2)

The Registration of Persons Act, Cap. 332

22 October 2026

To

The Executive Director
National Environment Management Authority of Uganda
Kampala

CONFIRMATION OF INFORMATION IN THE NATIONAL IDENTIFICATION REGISTER

Regulation 8(1) of the Registration of Persons Regulations, 2015, S.I. No. 67

Following an application made for **Confirmation of Information** in the National Identification Register, NIRA confirms that the facial image and corresponding bio-data set out below are recorded in the Register in respect of the person identified herein, as at the date of issue of this letter.

DATA SUBJECT



Surname: **WASSWA**
Given Name: **FRANCIS**
NIN: **CM00093106P55A**
Date of Birth: **01/05/2000**

CL/22/10/26/3882

R. Kisembo

Rosemary Kisembo
Executive Director

PLEASE NOTE:

1. You are required to take steps to **ascertain the authenticity of this letter** before relying on it by:
   a. *Scanning the QR code or visiting our website, **www.nira.go.ug**, under the section **"Verify Our Documents"** to confirm if this letter was electronically issued by NIRA; and*
2. If issued electronically, the link expires on first access or after seven (7) days. After expiry, a new application and payment are required.
3. Use the information only for the stated purpose and protect it from unauthorized access or misuse in compliance with the Data Protection and Privacy Act, Cap. 97.
4. NIRA will not be responsible for documents which are:
   a. Expired;
   b. Being used for a purpose other than the declared one; or
   c. Used by a party the letter is not addressed to; or
   d. Fraudulently issued or cannot be verified on **www.nira.go.ug** under the **"Verify Our Documents"** link; or
   e. Altered or reproduced.

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Appendix 1: Declaration form *(Print and complete this).*

I certify that the information provided in this application is true and accurate, that I meet the eligibility requirements of this call, and that the filled information is from the research proposal that has been approved by my supervisor and department.

Name of Applicant/Innovator	Signature	Date
WAISWA FRANCIS		15th/10/2026
Name of Supervisor	**Signature**	**Date**
MASERUKA BENEDICTO SSAJJABI		21/10/2026
Name of Head of Department	**Signature and stamp**	**Date**
SSEMWOGA PAUL	BUSITEMA UNIVERSITY DEPARTMENT OF CHEMICAL AND PROCESS ENGINEERING 21 OCT 2026 FACULTY OF ENGINEERING AND TECHNOLOGY	21/10/2026

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